# T9 — The Helmert-Fourier Transfer Matrix

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**Abstract**

**Situation.** T4 (Murillo Montero 2026, T4) introduced the **Helmert ladder** $\{h_k\}_{k=1}^{N-1}$ of integer orthogonal contrasts spanning the mean-zero hyperplane $H_N \subset \mathbb{R}^N$, and proved that unit normalization $e_k := h_k/\sqrt{k(k+1)}$ forces a quadratic field extension exactly at prime rungs, yielding the multiquadratic radical clock $K_N = \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime})$. T0 (T0, §3 and §5) introduced the **character basis** $\{\tilde{\chi}_c\}_{c=1}^{N-1}$ of $H_N \otimes \mathbb{C}$ — the discrete Fourier basis on $\mathbb{Z}/N$ minus the trivial mode — and used it to define the structured apex generator $\mathcal{M}_x(\beta, c)$ via the conductor lattice. T0 §5.5 computed the **Helmert-Fourier transfer matrix** at $N = 6$ in closed form, establishing Proposition 5.2:

$$\langle h_k, \chi_c \rangle = \frac{1-\omega^{kc}}{1-\omega^c} - k\omega^{kc}, \qquad \omega = e^{2\pi i/N},$$

and identified the radical-cyclotomic factorization of the matrix entries.

**Gap.** The closed-form entries are known (T0 Prop 5.2); but the full *structural* content of the transfer matrix as the apex's arithmetic Jacobian between the dimensional (Helmert/radical) and conductor (character/cyclotomic) axes was scoped in T0 §6.3 as a forward door — specifically, (a) the explicit factorization in three equivalent forms (rational polynomial, trigonometric, exponential), (b) the block structure over conductor packets $\{H_d\}_{d|N}$, (c) the row/column unitarity content, (d) general-$N$ worked examples beyond $N = 6$, and (e) the relationship to standard signal-processing transforms (DFT, DCT, DST, Hartley) and to Gauss sums.

**Action.** We establish the full structure theorem of the transfer matrix $U := (\langle e_k, \tilde{\chi}_c \rangle)_{k,c=1}^{N-1} \in U(N-1)$ (unitary $(N-1) \times (N-1)$ matrix over the compositum $K_N \cdot \mathbb{Q}(\zeta_N)$), with three equivalent closed forms for the entries, complete row and column structural analysis, the block decomposition over conductor packets (showing that $U$ is *not* generically block-diagonal but is *structured* by conductor classes via Gauss-sum-like identities), and worked examples at $N \in \{4, 5, 6, 7, 8\}$ covering prime, prime-power, and cross-prime cases.

**Result.** Seven core claims:

1. **Three closed forms.** $U_{kc}$ admits three equivalent expressions: a rational polynomial in $\omega^c = e^{2\pi ic/N}$, a trigonometric form with $\sin / \cos$ of $\pi kc/N$, and an exponential form mixing $e^{i\pi(k-1)c/N}$ phases.
2. **Radical-cyclotomic factorization.** Each entry $U_{kc} = R_k \cdot C_{kc}$ where $R_k = 1/\sqrt{Nk(k+1)} \in K_N$ is a radical depending only on $k$ (the Helmert axis), and $C_{kc}$ is a Laurent polynomial in $\omega^c$ with integer coefficients (the cyclotomic / conductor axis).
3. **Last-row uniformity.** $U_{N-1,c} = -\omega^{(N-1)c}/\sqrt{N-1}$ has constant modulus $1/\sqrt{N-1}$ for every $c$, with phase $-\omega^{(N-1)c}$. The last Helmert direction $e_{N-1}$ has uniform Fourier amplitude across all non-trivial characters. (Whether $e_{N-1}$ is the *unique* Helmert direction with this property — beyond worked examples at small $N$, where it is unique — is left open.)
4. **First-row sinusoidal.** $U_{1c} = (1-\omega^c)/\sqrt{2N}$ has modulus $\sqrt{2/N}|\sin(\pi c/N)|$, peaking at $c = N/2$ (when $N$ is even) — the Nyquist mode. The first Helmert direction is the discrete pair $(1, -1)$ at positions $0, 1$, and its Fourier content is sinusoidal in $c$.
5. **Block structure over conductors.** $U$ is *not* block-diagonal in the conductor-grouped basis (columns ordered by $\text{cond}(\chi_c)$), but the *column blocks* satisfy specific row-magnitude relations: the block at conductor $d \mid N$ has dimension $\varphi(d)$ and aggregates Gauss-sum-like content over the divisors of $d$.

6. **Unitarity in closed form.** Both row sums $\sum_c |U_{kc}|^2$ and column sums $\sum_k |U_{kc}|^2$ evaluate to 1 exactly via the identities

$$\sum_{c=1}^{N-1} |1-\omega^{kc}|^2 = 2(N-1), \qquad \sum_{c=1}^{N-1} (1-\cos(2\pi c/N)) = N.$$

7. **Worked examples at $N \in \{4,5,6,7,8\}$.** Explicit matrix forms over the compositum $K_N \cdot \mathbb{Q}(\zeta_N)$, exhibiting cyclic ($N = 5, 7$ prime), prime-power depth ($N = 4, 8$), and cross-prime interaction ($N = 6$) structures.

**Harvest.** The Helmert-Fourier transfer matrix $U$ is the categorical-simplex apex's **arithmetic Jacobian** between its two dual orthogonal bases of the mean-zero hyperplane: the integer-rational Helmert basis (carrying the dimensional / multiquadratic-radical clock structure of T4) and the cyclotomic character basis (carrying the conductor-packet / additive Fourier structure of T0/T3). Every apex computation that mixes the two axes — every transfer between dimensional and conductor content — passes through $U$. T9 closes the T0 §6.3 forward door at the entry-level, leaving the algebraic-group-theoretic content of $U$ (its action on the multiquadratic-cyclotomic compositum, its block-spectral structure, its relation to Walsh-Hadamard and DCT transforms) as continuing forward doors.

**Keywords:** Helmert ladder, character basis, change-of-basis matrix, unitary matrix, cyclotomic field, multiquadratic field, conductor packets, Gauss sums, DFT, DCT, Walsh-Hadamard, mean-zero hyperplane, finite Fourier analysis, arithmetic Jacobian.

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*Two orthogonal bases of $H_N$: one integer-rational (Helmert), one cyclotomic (character). Their transfer matrix is the apex's arithmetic Jacobian.*

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### Cross-references from prior papers

This paper builds on specific results from earlier papers of the T-series; we list the dependencies explicitly.

| Cited result | Used for |
|---|---|
| T0 §3.2 Definition 3.1 (apex generator $\mathcal{M}_x$) | Background; not used directly |
| T0 §3.3 Theorem 3.5 (closure diagonal) | Conductor selection rule context |
| T0 §4.2 (conductor packets $H_d$, Proposition 4.1 prime-power locality) | Conductor structure in §4 |
| T0 §5.2 Theorem 5.1 (multiquadratic radical clock, restating T4) | Multiquadratic field $K_N$ throughout |
| T0 §5.5 Proposition 5.2 (closed form $\langle h_k, \chi_c \rangle$) | Starting point — restated and elaborated as Theorem 2.1 |
| T0 §5.5 ($N = 6$ worked example) | §5.2 reference; not re-derived |
| T4 §3 (Helmert ladder construction, prime-rung growth) | Foundation of §1.1 |
| T8 §4.1 (Gauss sum $\tau(\psi, a)$) | §4.4 forward door |

No theorems from T5, T6, T7, T10 are used directly. T8 enters only through the Gauss-sum nomenclature in §4.4 (forward door).

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### Notation

We carry the notation of T0 and T4 and consolidate it for the transfer matrix.

| Symbol | Meaning |
|---|---|
| $N$ | Integer dimension, $\geq 2$. |
| $H_N$ | Mean-zero hyperplane $\{q \in \mathbb{R}^N : \sum q_i = 0\}$, dim $N-1$. |
| $h_k$ | Unnormalized Helmert vector $(1, \dots, 1, -k, 0, \dots, 0)$, $1 \leq k \leq N-1$. |
| $e_k$ | Unit Helmert vector $h_k / \sqrt{k(k+1)}$. |
| $\omega$ | Primitive $N$-th root of unity $e^{2\pi i/N}$. |
| $\chi_c$ | Additive character $\chi_c(s) = \omega^{cs} = e^{2\pi i cs/N}$ for $s \in \mathbb{Z}/N$. |
| $\tilde{\chi}_c$ | $L^2$-normalized character $\chi_c/\sqrt{N}$. |
| $U = (U_{kc})$ | Transfer matrix $U_{kc} := \langle e_k, \tilde{\chi}_c \rangle$, $k, c = 1, \dots, N-1$. |
| $K_N$ | Multiquadratic field $\mathbb{Q}(\sqrt{p} : p \leq N,\ p$ prime) (T4). |
| $\mathbb{Q}(\zeta_N)$ | $N$-th cyclotomic field $\mathbb{Q}(\omega)$. |
| $K_N \cdot \mathbb{Q}(\zeta_N)$ | Compositum of $K_N$ and $\mathbb{Q}(\zeta_N)$ inside $\mathbb{C}$. |

| Symbol | Meaning |
|---|---|
| $\mathrm{cond}(\chi_c)$ | Conductor $N/\gcd(c,N)$ of $\chi_c$. |
| $H_d$ | Conductor packet $\mathrm{span}\{\chi_c : \mathrm{cond}(\chi_c) = d\}$, $d \mid N$, $d > 1$. |
| $\varphi$ | Euler totient function. |
| $\tau(\psi, a)$ | Gauss sum $\sum_{s\in(\mathbb{Z}/N)^\times} \psi(s)\chi_a(s)$ (T8 notation). |

**Convention.** All vectors are column vectors when concrete; inner products on $\mathbb{C}^N$ are Hermitian: $\langle u, v\rangle = \sum_i \bar{u}_i v_i$. The Helmert vectors $h_k$ are real, so $\langle h_k, \chi_c\rangle = \sum_s (h_k)_s \chi_c(s)$ without conjugation. The transfer matrix $U$ is unitary: $U^*U = UU^* = I_{N-1}$.

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## 1. The two bases of $H_N$

### 1.1 The Helmert basis (integer / rational)

The mean-zero hyperplane $H_N = \{q \in \mathbb{R}^N : \sum_i q_i = 0\}$ has dimension $N-1$. The **Helmert basis** (Helmert 1876; Lancaster 1965) is the canonical integer orthogonal basis of $H_N$ given by the contrasts

$$h_k = (\underbrace{1, 1, \ldots, 1, -k}_{k \text{ entries}}, \underbrace{0, \ldots, 0}_{N-k-1 \text{ entries}}) \in \mathbb{R}^N, \qquad k = 1, 2, \ldots, N-1.$$

The squared norms are $\|h_k\|^2 = k + k^2 = k(k+1)$, and the orthogonality relations $\langle h_j, h_k\rangle = 0$ for $j \neq k$ are routine to verify.

The **unit Helmert vectors** are $e_k := h_k/\sqrt{k(k+1)}$, forming an orthonormal basis of $H_N$.

The Helmert basis carries two distinctive structural features: - **Integer entries.** The $h_k$ have entries in $\{-k, 0, 1\}$, an integer rational basis of $H_N$ (without normalization). - **Multiquadratic radical normalization.** The unit normalizers $\sqrt{k(k+1)}$ accumulate, across the dimensional axis $N$, into the multiquadratic field $K_N = \mathbb{Q}(\sqrt{p} : p \leq N,\ p \text{ prime})$ (T4 Theorem 3, restated in T0 §5.2 Theorem 5.1).

### 1.2 The character basis (cyclotomic / complex)

The mean-zero hyperplane extends to $H_N \otimes \mathbb{C} \subset \mathbb{C}^N$, with the **character basis** $\{\chi_c\}_{c=1}^{N-1}$ where $\chi_c(s) = \omega^{cs} = e^{2\pi i cs/N}$ viewed as a vector in $\mathbb{C}^N$ with components $(\chi_c(s))_{s=0}^{N-1}$. Each non-trivial character $\chi_c$ has $\sum_s \chi_c(s) = 0$, hence lies in $H_N \otimes \mathbb{C}$. The $N-1$ non-trivial characters are linearly independent and span $H_N \otimes \mathbb{C}$ over $\mathbb{C}$.

Standard norm computation: $\|\chi_c\|^2 = \sum_s |\chi_c(s)|^2 = N$, so the $L^2$-normalized characters are $\tilde{\chi}_c := \chi_c/\sqrt{N}$, forming an orthonormal basis of $H_N \otimes \mathbb{C}$.

The character basis carries: - **Complex unitary entries.** $\tilde{\chi}_c$ has entries in $\mathbb{Q}(\zeta_N)\cdot(1/\sqrt{N})$, where $\mathbb{Q}(\zeta_N) = \mathbb{Q}(\omega)$ is the $N$-th cyclotomic field. - **Conductor packet structure.** Characters group by conductor $\mathrm{cond}(\chi_c) = N/\gcd(c,N)$ into conductor packets $H_d \subset H_N \otimes \mathbb{C}$ of dimensions $\varphi(d)$ (T0 §4.2).

### 1.3 The transfer matrix

**Definition 1.1 (Helmert-Fourier transfer matrix).** The **transfer matrix** between the Helmert and character bases of $H_N \otimes \mathbb{C}$ is

$$U = (U_{kc})_{k,c=1}^{N-1} \in \mathbb{C}^{(N-1)\times(N-1)}, \qquad U_{kc} := \langle e_k, \tilde{\chi}_c \rangle = \frac{1}{\sqrt{Nk(k+1)}} \langle h_k, \chi_c \rangle.$$

The unit Helmert vector $e_k$ has Fourier expansion

$$e_k = \sum_{c=1}^{N-1} \overline{U_{kc}}\, \tilde{\chi}_c,$$

and conversely, the normalized character $\tilde{\chi}_c$ has Helmert expansion

$$\tilde{\chi}_c = \sum_{k=1}^{N-1} U_{kc}\, e_k.$$

**Proposition 1.2 (Unitarity).** $U \in U(N-1)$; equivalently, $U^*U = UU^* = I_{N-1}$.

**Proof.** Both $\{e_k\}$ and $\{\tilde{\chi}_c\}$ are orthonormal bases of $H_N \otimes \mathbb{C}$ with respect to the Hermitian inner product on $\mathbb{C}^N$ restricted to $H_N \otimes \mathbb{C}$. The change-of-basis matrix between two orthonormal bases is unitary. $\square$

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## 2. Three closed forms for the entries

### 2.1 Rational polynomial form

**Theorem 2.1 (Rational polynomial form).** For $1 \le k \le N-1$ and $c \in \{1, \dots, N-1\}$,

$$\langle h_k, \chi_c \rangle = \frac{1-\omega^{kc}}{1-\omega^c} - k\omega^{kc} = \frac{1-(k+1)\omega^{kc}+k\omega^{(k+1)c}}{1-\omega^c}.$$

Equivalently, $\langle h_k, \chi_c \rangle$ is a Laurent polynomial in $\omega^c$ with integer coefficients (after clearing the $1-\omega^c$ denominator), of degree $k+1$ in $\omega^c$ with leading coefficient $k$ and constant term 1.

**Proof.** The first equality is T0 Proposition 5.2, restated; computed directly from $h_k = (1, \dots, 1, -k, 0, \dots, 0)$:

$$\langle h_k, \chi_c \rangle = \sum_{j=0}^{k-1} \omega^{cj} - k\omega^{kc} = \frac{1-\omega^{kc}}{1-\omega^c} - k\omega^{kc}.$$

The second equality follows by clearing denominators:

$$\frac{1-\omega^{kc}}{1-\omega^c} - k\omega^{kc} = \frac{(1-\omega^{kc}) - k\omega^{kc}(1-\omega^c)}{1-\omega^c} = \frac{1-(k+1)\omega^{kc}+k\omega^{(k+1)c}}{1-\omega^c}.$$

The numerator $P_k(z) := 1-(k+1)z^k + kz^{k+1}$ is a polynomial in $z = \omega^c$ with integer coefficients. Evaluating at $z=1$: $P_k(1) = 1-(k+1)+k = 0$, so $(1-z) \mid P_k(z)$; the quotient $P_k(z)/(1-z)$ is a polynomial in $z$ of degree $k$, with integer coefficients, agreeing with the geometric-sum expression $\sum_{j=0}^{k-1} z^j - kz^k$. $\square$

### 2.2 Trigonometric form

**Theorem 2.2 (Trigonometric form).** Setting $\theta_c := \pi c/N \in (0,\pi)$ for $c \in \{1,\dots,N-1\}$,

$$\langle h_k, \chi_c\rangle = e^{i(k-1)\theta_c}\frac{\sin(k\theta_c)}{\sin\theta_c} - ke^{2ik\theta_c}.$$

**Proof.** Using $1-\omega^a = 1-e^{2ia\theta_c} = -2ie^{ia\theta_c}\sin(a\theta_c)$,

$$\frac{1-\omega^{kc}}{1-\omega^c} = \frac{-2ie^{ik\theta_c}\sin(k\theta_c)}{-2ie^{i\theta_c}\sin\theta_c} = e^{i(k-1)\theta_c}\frac{\sin(k\theta_c)}{\sin\theta_c}.$$

Substituting into Theorem 2.1 gives the stated form. □

The trigonometric form makes manifest the **sinusoidal envelope** $\sin(k\theta_c)/\sin\theta_c$ characteristic of Dirichlet-kernel-like structures in classical Fourier analysis.

### 2.3 Exponential form

**Theorem 2.3 (Symmetric exponential form).** Factoring out the geometric center of the phase content,

$$\langle h_k, \chi_c\rangle = e^{ik\theta_c}\Big[e^{-i\theta_c}\frac{\sin(k\theta_c)}{\sin\theta_c} - ke^{ik\theta_c}\Big].$$

This identifies the **phase factor** $e^{ik\theta_c} = \omega^{kc/2}$ (where the half-power is the natural symmetric branch of $\omega^c$) and an **amplitude bracket** that is asymptotically dominated by the $\sin(k\theta_c)/\sin\theta_c$ term for small $k$ and by $-ke^{ik\theta_c}$ for $k$ near $N-1$.

**Proof.** Direct factorization of Theorem 2.2. □

### 2.4 Entries of $U$ in the radical-cyclotomic factorization

Putting the three forms together with the unit Helmert normalization:

**Corollary 2.4 (Transfer matrix entries).** For $k,c \in \{1,\dots,N-1\}$,

$$\boxed{U_{kc} = \frac{1}{\sqrt{Nk(k+1)}}\cdot\left[e^{i(k-1)\theta_c}\sin(k\theta_c)/\sin\theta_c - ke^{2ik\theta_c}\right] = \frac{1}{\sqrt{Nk(k+1)}}\cdot\frac{1-(k+1)\omega^{kc}+k\omega^{(k+1)c}}{1-\omega^c},}$$

where $\theta_c = \pi c/N$ and $\omega = e^{2\pi i/N}$. The factor $1/\sqrt{Nk(k+1)} \in K_N$ is a **radical** depending only on $k$ (the Helmert axis). The rational-polynomial form on the right makes manifest that the cyclotomic content lives in $\mathbb{Q}(\zeta_N) = \mathbb{Q}(\omega)$ (Theorem 2.1); the trigonometric form on the left uses half-roots $\zeta_{2N}$ as a representation, but the actual value of the bracket lies in $\mathbb{Q}(\zeta_N)$ after the cancellation $e^{i(k-1)\theta_c}\sin(k\theta_c)/\sin\theta_c \in \mathbb{Q}(\zeta_N)$ that is forced by equality with the rational form. Therefore each entry $U_{kc}$ lives in the compositum $K_N\cdot\mathbb{Q}(\zeta_N)$.

The compositum $K_N\cdot\mathbb{Q}(\zeta_N)$ is the **arithmetic field of the transfer matrix** at level $N$. For $N=6$ this is $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5},\zeta_6)$ of degree 16 over $\mathbb{Q}$ (cf. T0 §5.5).

**Remark 2.5 (Bilateral structure of the radical factor).** The radical $R_k$ further splits as

$$R_k = \frac{1}{\sqrt{Nk(k+1)}} = \frac{1}{\sqrt{N}}\cdot\frac{1}{\sqrt{k(k+1)}},$$

exhibiting two structurally distinct contributions:

- $1/\sqrt{k(k+1)}$ is the **unit-Helmert normalizer** of the $k$-th contrast (T4 §3 / T0 §5.2). By T4 Theorem 3.1 (restated as T0 Theorem 5.1), these normalizers collectively generate the multiquadratic radical field $K_N = \mathbb{Q}(\sqrt{p} : p \leq N)$ as $k$ ranges over $\{1, \ldots, N-1\}$, growing properly over $K_{N-1}$ precisely at prime rungs.
- $1/\sqrt{N}$ is the **level-$N$ ambient**, arising from the $L^2$-normalization $\tilde{\chi}_c = \chi_c/\sqrt{N}$ of the character basis (§1.2). It lies in $K_N$ since the prime factorization of $N$ involves only primes $p \mid N \leq N$, all of which contribute $\sqrt{p} \in K_N$ as generators.

Both factors live in $K_N$ — the radical $R_k$ is built from T4's Helmert-axis arithmetic together with the character-basis ambient, and the entire entry $U_{kc}$ is then $R_k$ times the cyclotomic conductor data $C_{kc} \in \mathbb{Q}(\zeta_N)$. This is the bilateral structure of the transfer matrix: T4 radical clock × T0/T3 character cyclotomic, intertwined at the entry level.

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## 3. Row and column structure

The transfer matrix $U \in U(N-1)$ has specific structural features visible at the row and column level.

### 3.1 The last row: uniform Fourier amplitude

**Proposition 3.1 (Last row uniformity).** $U_{N-1,c} = -\omega^{(N-1)c}/\sqrt{N-1}$ for every $c \in \{1, \ldots, N-1\}$. The modulus is the constant $1/\sqrt{N-1}$ across all $c$.

**Proof.** $h_{N-1} = (1, 1, \ldots, 1, -(N-1))$ has $N-1$ ones and $-(N-1)$ at position $N-1$. Therefore

$$\langle h_{N-1}, \chi_c \rangle = \sum_{s=0}^{N-2} \omega^{cs} - (N-1)\omega^{(N-1)c}.$$

For $c \neq 0$, $\sum_{s=0}^{N-1} \omega^{cs} = 0$ (additive character orthogonality), so $\sum_{s=0}^{N-2} \omega^{cs} = -\omega^{(N-1)c}$. Substituting,

$$\langle h_{N-1}, \chi_c \rangle = -\omega^{(N-1)c} - (N-1)\omega^{(N-1)c} = -N\omega^{(N-1)c}.$$

Dividing by $\sqrt{N \cdot (N-1) \cdot N} = N\sqrt{N-1}$,

$$U_{N-1,c} = \frac{-N\omega^{(N-1)c}}{N\sqrt{N-1}} = -\frac{\omega^{(N-1)c}}{\sqrt{N-1}}. \qquad \square$$

**Interpretation.** The last Helmert direction $e_{N-1} = h_{N-1}/\sqrt{N(N-1)}$ — the maximally "rear-loaded" contrast, with weight $-(N-1)$ at the last position — has **uniform Fourier amplitude** across all non-trivial characters. In the worked examples of §5 ($N \leq 8$), $e_{N-1}$ is the *unique* Helmert direction with this property; whether uniqueness holds for all $N$ is left open.

### 3.2 The first row: sinusoidal Fourier amplitude

**Proposition 3.2 (First row sinusoidal).** $U_{1c} = (1 - \omega^c)/\sqrt{2N}$, with $|U_{1c}| = \sqrt{2/N} \cdot |\sin(\pi c/N)| = \sqrt{2/N}\sin\theta_c$ for $c \in \{1, \ldots, N-1\}$.

**Proof.** $h_1 = (1, -1, 0, \ldots, 0)$, so $\langle h_1, \chi_c \rangle = 1 - \omega^c$. Dividing by $\sqrt{N \cdot 2} = \sqrt{2N}$ gives $U_{1c}$. The modulus computation: $|1 - \omega^c|^2 = (1 - \cos 2\theta_c)^2 + \sin^2 2\theta_c = 2 - 2\cos 2\theta_c = 4\sin^2\theta_c$, so $|1 - \omega^c| = 2|\sin\theta_c|$ and $|U_{1c}| = 2|\sin\theta_c|/\sqrt{2N} = \sqrt{2/N}\sin\theta_c$ (for $\theta_c \in (0, \pi)$, $\sin\theta_c > 0$). $\square$

**Interpretation.** The first Helmert direction $e_1 = (1, -1, 0, \dots, 0)/\sqrt{2}$ — the simplest discrete contrast — has a **sinusoidal Fourier amplitude** in $c$, peaking at $c = N/2$ (the Nyquist mode, when $N$ is even) with peak amplitude $\sqrt{2/N}$. This is the **discrete sine-transform signature** of the first Helmert contrast.

### 3.3 Unitarity in closed form

The row and column sums of $|U|^2$ both evaluate to 1 by unitarity, but the algebraic identities involved are worth recording.

**Proposition 3.3 (Row unitarity).** $\sum_{c=1}^{N-1} |U_{kc}|^2 = 1$ for every $k \in \{1, \dots, N-1\}$.

**Proof.** $e_k$ is a unit vector in $H_N \otimes \mathbb{C}$ (Helmert orthonormality), and $\{\tilde{\chi}_c\}_{c=1}^{N-1}$ is an orthonormal basis of $H_N \otimes \mathbb{C}$ (Definition 1.1). Parseval's identity gives

$$\sum_{c=1}^{N-1} |\langle e_k, \tilde{\chi}_c \rangle|^2 = \|e_k\|^2 = 1,$$

which is the claim, since $U_{kc} = \langle e_k, \tilde{\chi}_c \rangle$. □

**Corollary 3.3a (Trigonometric identity at $k = 1$).** Parseval's identity applied to $e_1$ yields the closed-form identity

$$\sum_{c=1}^{N-1} \sin^2(\pi c/N) = \frac{N}{2}.$$

**Proof.** $|U_{1c}|^2 = (2/N)\sin^2(\pi c/N)$ (Proposition 3.2), so Proposition 3.3 at $k = 1$ gives $(2/N)\sum_{c=1}^{N-1} \sin^2(\pi c/N) = 1$, equivalently $\sum_c \sin^2(\pi c/N) = N/2$. (This identity also follows from $\sin^2 x = (1 - \cos 2x)/2$ and the additive-character identity $\sum_{c=0}^{N-1} \cos(2\pi c/N) = 0$.) □

**Proposition 3.4 (Column unitarity).** $\sum_{k=1}^{N-1} |U_{kc}|^2 = 1$ for every $c \in \{1, \dots, N-1\}$.

**Proof.** Parallel to Proposition 3.3: $\tilde{\chi}_c$ is a unit vector and $\{e_k\}$ is an orthonormal basis, so $\sum_k |\langle \tilde{\chi}_c, e_k \rangle|^2 = \|\tilde{\chi}_c\|^2 = 1$. Since $\langle \tilde{\chi}_c, e_k \rangle = \overline{U_{kc}}$, the squared moduli are equal. □

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## 4. The conductor-packet block structure

### 4.1 Conductor reordering of columns

The character basis $\{\tilde{\chi}_c\}_{c=1}^{N-1}$ partitions naturally by conductor:

$$\{1, \dots, N-1\} = \bigsqcup_{d|N,\, d>1} C_d, \qquad C_d := \{c \in \{1, \dots, N-1\} : \mathrm{cond}(\chi_c) = d\}.$$

By T0 §4.2, $|C_d| = \varphi(d)$ for each divisor $d > 1$ of $N$. Reordering columns of $U$ by conductor groups produces a **block-columnar** structure

$$U = [U^{(d_1)} \,|\, U^{(d_2)} \,|\, \cdots \,|\, U^{(d_r)}],$$

where the divisors $d_1 < d_2 < \cdots < d_r$ are the divisors of $N$ greater than 1, and $U^{(d)} \in \mathbb{C}^{(N-1)\times\varphi(d)}$ is the column-block of $U$ indexed by characters of conductor $d$.

### 4.2 Block structure is *not* block-diagonal in general

A natural question: does the Helmert basis admit a parallel partition by conductor, producing a block-diagonal $U$?

**Proposition 4.1 (Non-block-diagonality at the first Helmert vector).** For every $N \geq 3$, no permutation of the Helmert basis $\{e_k\}_{k=1}^{N-1}$ makes $U$ block-diagonal with respect to the conductor partition of columns. In particular, the first Helmert vector $h_1 = (1, -1, 0, \ldots, 0)$ has non-zero Fourier content in *every* conductor packet $H_d$ for $d \mid N$, $d > 1$.

**Proof.** Block-diagonality of $U$ (under some permutation of rows) would require each Helmert vector $h_k$ to be supported on a single conductor packet $H_d$ — i.e., $\langle h_k, \chi_c \rangle = 0$ for every $c$ with $\mathrm{cond}(\chi_c) \neq d$.

For $h_1$: $\langle h_1, \chi_c \rangle = 1 - \omega^c$, which vanishes only when $\omega^c = 1$, i.e., $c \equiv 0 \pmod{N}$. Since $c \in \{1, \ldots, N-1\}$, $\omega^c \neq 1$ for every non-trivial character, so $\langle h_1, \chi_c \rangle \neq 0$ for *every* $c \neq 0$ — in particular, for every $c$ in *every* conductor packet $H_d$ ($d \mid N$, $d > 1$). Hence $h_1$ has non-zero content in every conductor packet simultaneously and cannot be assigned to a single packet under any permutation. □

The argument generalizes: $h_k$ has zero Fourier content at $\chi_c$ iff $1 - (k+1)\omega^{kc} + k\omega^{(k+1)c} = 0$ (Theorem 2.1 numerator), an equation in $\omega^c$ of degree $k+1$ with at most $k+1$ solutions; for $k+1 < N-1$ this leaves $\geq 1$ non-zero values of $c$ on which $h_k$ has non-zero content per conductor packet, generically preventing assignment to a single packet.

So $U$ is structured by conductor columns but does *not* generically decompose into a block-diagonal form. The transfer between the Helmert and character bases genuinely mixes conductor packets.

### 4.3 Column-block structural relations

While not block-diagonal, the conductor-column blocks $U^{(d)}$ satisfy specific structural relations:

**Proposition 4.2 (Column-block norms).** For each divisor $d \mid N$ with $d > 1$,

$$\sum_{c \in C_d} |U_{kc}|^2 = \frac{1}{Nk(k+1)} \sum_{c \in C_d} |\langle h_k, \chi_c \rangle|^2$$

is the **conductor-$d$ content** of the unit Helmert vector $e_k$. The total $\sum_{d|N, d>1} \sum_{c \in C_d} |U_{kc}|^2 = \sum_{c=1}^{N-1} |U_{kc}|^2 = 1$ partitions the unit norm of $e_k$ across the conductor packets.

This gives a **conductor-spectral decomposition** of each Helmert basis vector: how much of $e_k$ lives at conductor $d$. Computing this in closed form via the rational polynomial expression of Theorem 2.1 is a finite combinatorial problem in each case.

### 4.4 Where Gauss sums enter

The conductor blocks $U^{(d)}$ involve partial geometric sums $\sum_{j=0}^{k-1} \omega^{cj}$ over additive characters of conductor $d$, but these are *not* themselves Gauss sums in the classical sense: Gauss sums (T8 §4.1) are sums over $(\mathbb{Z}/N)^\times$ of a *multiplicative-character* weighting times an additive character, of the form $\tau(\psi, a) = \sum_{s \in (\mathbb{Z}/N)^\times} \psi(s)\chi_a(s)$.

The proper connection between T9's transfer matrix and T8's Gauss sums goes via the **conductor-$N$ block**: when $d = N$ (i.e., $\chi_c$ has conductor $N$, equivalently $\gcd(c, N) = 1$), the character $\chi_c$

restricted to units of $(\mathbb{Z}/N)^\times$ is a *primitive additive character*, and the row-aggregated content $\sum_{c\in C_N}(\text{multiplicative-character weight})\cdot\langle h_k,\chi_c\rangle$ becomes a Gauss sum proper. Precise statements of this aggregation — including which multiplicative characters $\psi$ produce non-trivial Gauss-sum identities and how the Helmert ladder contributes structurally — are forward doors (§7.4) connecting T9 to T8's bilateral Pontryagin theory.

The conductor-$d$ block $U^{(d)}$ for $d < N$ involves additive characters of smaller conductor, lifted from $\mathbb{Z}/d$ to $\mathbb{Z}/N$; its Gauss-sum content lives at level $d$, not $N$. The interaction between conductor-$d$ blocks and Gauss sums at level $d$ requires the depth-filtration analysis of T0 §4.2 generalized to the bilateral setting.

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## 5. Worked examples

### 5.1 $N = 4$: smallest prime-power-depth case

For $N = 4$: $\omega = e^{i\pi/2} = i$. Divisors of 4 greater than 1: $\{2, 4\}$. - $C_2 = \{2\}$ (conductor 2, $\chi_2(s) = (-1)^s$). - $C_4 = \{1, 3\}$ (conductor 4).

Helmert vectors: $h_1 = (1, -1, 0, 0)$, $\|h_1\| = \sqrt{2}$. $h_2 = (1, 1, -2, 0)$, $\|h_2\| = \sqrt{6}$. $h_3 = (1, 1, 1, -3)$, $\|h_3\| = \sqrt{12} = 2\sqrt{3}$.

Direct computation of $\langle h_k, \chi_c\rangle = \sum_{j=0}^{k-1} i^{cj} - ki^{kc}$ for $k, c \in \{1, 2, 3\}$ yields

| $k\backslash c$ | $c = 1$ | $c = 2$ | $c = 3$ |
|---|---|---|---|
| $k = 1$ | $1 - i$ | $2$ | $1 + i$ |
| $k = 2$ | $3 + i$ | $-2$ | $3 - i$ |
| $k = 3$ | $4i$ | $4$ | $-4i$ |

Normalizing by $\sqrt{Nk(k+1)} = \sqrt{4k(k+1)}$ ($k = 1 : \sqrt{8} = 2\sqrt{2}$; $k = 2 : \sqrt{24} = 2\sqrt{6}$; $k = 3 : \sqrt{48} = 4\sqrt{3}$) gives the transfer matrix:

$$U_4 = \begin{pmatrix} \frac{1-i}{2\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1+i}{2\sqrt{2}} \\ \frac{3+i}{2\sqrt{6}} & \frac{-1}{\sqrt{6}} & \frac{3-i}{2\sqrt{6}} \\ \frac{i}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{-i}{\sqrt{3}} \end{pmatrix}.$$

Observations: - Last row $k = 3$: constant modulus $1/\sqrt{3}$ in agreement with Proposition 3.1 ($1/\sqrt{N-1} = 1/\sqrt{3}$). - First row $k = 1$: moduli $(|1 - i|, |2|, |1 + i|)/(2\sqrt{2}) = (\sqrt{2}, 2, \sqrt{2})/(2\sqrt{2}) = (1/2, 1/\sqrt{2}, 1/2)$, consistent with Proposition 3.2 sinusoidal form $\sqrt{2/N}\sin(\pi c/N)$ evaluated at $\theta = \pi/4, \pi/2, 3\pi/4$. - Row-unitarity check $k = 1$: $|U_{1c}|^2$ sums to $1/4 + 1/2 + 1/4 = 1$.

The arithmetic field of $U_4$ is $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6}, i) = \mathbb{Q}(\sqrt{2}, \sqrt{3}, i) = K_4 \cdot \mathbb{Q}(\zeta_4)$ — degree 8 over $\mathbb{Q}$.

### 5.2 $N = 6$: cross-prime case (from T0 §5.5)

Already worked in T0 §5.5. We do not repeat the full computation here; the key data: - $\omega = e^{i\pi/3}$, $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$. - Conductor classes: $C_2 = \{3\}$, $C_3 = \{2, 4\}$, $C_6 = \{1, 5\}$. - Arithmetic field: $K_6 \cdot \mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \zeta_6)$, degree 16. - The $c = 3$ column ($H_2$ packet, real characters) computed in T0 §5.5: $U_{\cdot,3} = (1/\sqrt{3}, -1/3, \sqrt{2}/3, -2/\sqrt{30}, 1/\sqrt{5})^\top$.

### 5.3 $N = 8$: prime-power depth, non-cyclic units

For $N = 8$: $\omega = e^{i\pi/4} = (1 + i)/\sqrt{2}$. Divisors of 8 greater than 1: $\{2, 4, 8\}$, with conductor classes

$$C_2 = \{4\},\ |C_2| = \varphi(2) = 1; \qquad C_4 = \{2, 6\},\ |C_4| = \varphi(4) = 2; \qquad C_8 = \{1, 3, 5, 7\},\ |C_8| = \varphi(8) = 4,$$

totaling $1 + 2 + 4 = 7 = N - 1$.

**Arithmetic field.** $K_8 = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7})$, degree $2^4 = 16$ (one quadratic per prime $\leq 8$); $\mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2}, i)$, degree $\varphi(8) = 4$; the intersection $K_8 \cap \mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2})$ (since $K_8$ is totally real, excluding $i$; and the only quadratic in $\mathbb{Q}(\zeta_8)$ that lies in $K_8$ is $\mathbb{Q}(\sqrt{2})$). Hence

$$[K_8 \cdot \mathbb{Q}(\zeta_8) : \mathbb{Q}] = \frac{16 \cdot 4}{2} = 32.$$

**Helmert ladder.** The seven contrasts $h_1, \ldots, h_7$ have unit normalizers $\sqrt{2}, \sqrt{6}, \sqrt{12} = 2\sqrt{3}, \sqrt{20} = 2\sqrt{5}, \sqrt{30}, \sqrt{42}, \sqrt{56} = 2\sqrt{14}$, drawing radicals from the primes $\{2, 3, 5, 7\} \cap K_8$.

**Conductor structure.** Three packets of dimensions $1, 2, 4$, exhibiting **prime-power depth** $\{p, p^2, p^3\}$ for $p = 2$ (T0 Proposition 4.1).

**Last and first rows from §3.** $U_{7,c} = -\omega^{7c}/\sqrt{7}$ (constant modulus $1/\sqrt{7}$); $U_{1,c} = (1 - \omega^c)/4$ (modulus $|\sin(\pi c/8)|/2$, peaking at $c = 4$ where $|U_{1,4}| = 1/2$).

**Worked column at $c = 4$ (the $C_2$ packet).** The single column with $c = 4$ corresponds to $\chi_4(s) = (-1)^s$, real-valued. By Theorem 2.1 specialized to $\omega^c = -1$, using $\sum_{j=0}^{k-1}(-1)^j = (1 - (-1)^k)/2$,

$$\langle h_k, \chi_4 \rangle = \frac{1 - (-1)^k}{2} - k(-1)^k,$$

evaluating to the integer sequence $(2, -2, 4, -4, 6, -6, 8)$ for $k = 1, \ldots, 7$. Dividing by $\sqrt{Nk(k+1)} = \sqrt{8k(k+1)}$:

$$U_{\cdot,4} = \Big(\frac{1}{2}, -\frac{1}{2\sqrt{3}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{10}}, \frac{3}{2\sqrt{15}}, -\frac{3}{2\sqrt{21}}, \frac{1}{\sqrt{7}}\Big)^\top.$$

Column unitarity check (column $c = 4$): $\sum_{k=1}^{7} |U_{k,4}|^2$. Computing each term and putting over common denominator 420:

$$\frac{1}{4} = \frac{105}{420},\ \frac{1}{12} = \frac{35}{420},\ \frac{1}{6} = \frac{70}{420},\ \frac{1}{10} = \frac{42}{420},\ \frac{9}{60} = \frac{63}{420},\ \frac{9}{84} = \frac{45}{420},\ \frac{1}{7} = \frac{60}{420},$$

summing to $105 + 35 + 70 + 42 + 63 + 45 + 60 = 420$, i.e., $420/420 = 1$, confirming Proposition 3.4.

This column lives entirely in the real subfield $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{14}) \cdot \mathbb{Q} = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}) = K_8$ (no cyclotomic content, since $\chi_4$ is the real Dirichlet character). The cyclotomic content $\mathbb{Q}(\zeta_8)$ enters only in the higher-conductor columns $C_4$ and $C_8$.

The full $7 \times 7$ transfer matrix at $N = 8$ has the $c = 4$ column above; the $C_4$ block ($c = 2, 6$) requires $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$ content; and the $C_8$ block ($c = 1, 3, 5, 7$) requires the full $\mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2}, i)$ content. We do not compute the remaining 48 entries explicitly.

### 5.4 $N = 5, 7$: prime $N$ (single conductor)

For prime $N$: every non-trivial character has conductor $N$ (since $\gcd(c, N) = 1$ for $c \in \{1, \ldots, N-1\}$). The conductor packet structure is trivial — single packet $H_N$ of dimension $N-1$. The transfer matrix $U$ has no conductor-block-column refinement (only one block, of full size); structurally, $U$ at prime $N$ is "all one packet."

$N = 5$. $K_5 = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$, degree 8. $\mathbb{Q}(\zeta_5)$ has degree $\varphi(5) = 4$ and is a CM field whose maximal real subfield is the unique quadratic subfield $\mathbb{Q}(\sqrt{5})$ (since $5 \equiv 1 \pmod 4$, the Gauss-sum identity $\tau(\chi)^2 = 5$ for the quadratic Dirichlet character mod 5 places $\sqrt{5}$ in $\mathbb{Q}(\zeta_5)$). Hence the intersection $K_5 \cap \mathbb{Q}(\zeta_5) = \mathbb{Q}(\sqrt{5})$ has degree 2, and the compositum

$$[K_5 \cdot \mathbb{Q}(\zeta_5) : \mathbb{Q}] = \frac{8 \cdot 4}{2} = 16.$$

$N = 7$. $K_7 = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7})$, degree 16. $\mathbb{Q}(\zeta_7)$ has degree $\varphi(7) = 6$ and is a CM field. Its maximal real subfield $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ is a cyclic cubic field of degree 3 over $\mathbb{Q}$ with discriminant 49; this cubic does *not* contain $\sqrt{p}$ for any prime $p \leq 7$, since a cyclic cubic over $\mathbb{Q}$ cannot contain a quadratic extension. Since $K_7$ is totally real, $K_7 \cap \mathbb{Q}(\zeta_7) \subseteq K_7 \cap (\mathbb{Q}(\zeta_7) \cap \mathbb{R}) = K_7 \cap \mathbb{Q}(\zeta_7 + \zeta_7^{-1}) = \mathbb{Q}$ (the cubic real subfield meets the multiquadratic $K_7$ trivially by the degree argument). Hence

$$[K_7 \cdot \mathbb{Q}(\zeta_7) : \mathbb{Q}] = 16 \cdot 6 = 96.$$

### 5.5 Comparison table across $N$

| $N$ | Type | Conductor blocks (sizes) | $K_N$ degree | $\mathbb{Q}(\zeta_N)$ degree | Intersection degree | Compositum degree |
|---|---|---|---|---|---|---|
| 4 | $p^2$, $p = 2$ | $H_2, H_4 =$ $(1, 2)$ | 4 | 2 | 1 ($\mathbb{Q}$) | 8 |
| 5 | prime | $H_5 = (4)$ | 8 | 4 | 2 ($\mathbb{Q}(\sqrt{5})$) | 16 |
| 6 | $2 \cdot 3$ | $H_2, H_3, H_6 =$ $(1, 2, 2)$ | 8 | 2 | 1 ($\mathbb{Q}$) | 16 |
| 7 | prime | $H_7 = (6)$ | 16 | 6 | 1 ($\mathbb{Q}$) | 96 |
| 8 | $p^3$, $p = 2$ | $H_2, H_4, H_8 =$ $(1, 2, 4)$ | 16 | 4 | 2 ($\mathbb{Q}(\sqrt{2})$) | 32 |
| 30 | $2 \cdot 3 \cdot 5$ | $H_2, H_3, H_5, H_6, H_{10}, H_{15}, H_{30} =$ $(1, 2, 4, 2, 4, 8, 8)$ | 1024 | 8 | 2 ($\mathbb{Q}(\sqrt{5})$) | 4096 |

The arithmetic complexity of the transfer matrix grows quickly with $N$, but the structural decomposition (conductor blocks, radical-cyclotomic factorization, Helmert row structure) is uniform across all $N$.

### 5.6 $N = 30$: three-prime squarefree structural sample

At the squarefree-composite anchor $N = 30 = 2 \cdot 3 \cdot 5$ — the smallest $N$ with three distinct prime factors — the transfer matrix is $29 \times 29$, with rich conductor and radical content. We do not compute all 841 entries explicitly, but record the structural ingredients.

**Radical content.** The $L^2$-normalizers $\sqrt{30k(k+1)}$ for $k = 1, \ldots, 29$ generate radicals $\sqrt{\mathrm{sf}(30k(k+1))}$ where sf is the squarefree kernel. By T4 Theorem 5.1 / T0 §5.2, these radicals collectively generate $K_{30} = \mathbb{Q}(\sqrt{p} : p \leq 30,\ p \text{ prime}) = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{11}, \sqrt{13}, \sqrt{17}, \sqrt{19}, \sqrt{23}, \sqrt{29})$, a multiquadratic field of degree $2^{10} = 1024$ over $\mathbb{Q}$. Each prime $p \leq 30$ enters at the rung $k = p - 1$ via the relation $k(k+1) = (p-1)p$. A sample of representative new-prime rungs:

| $k$ | $30k(k+1)$ | $\mathrm{sf}(30k(k+1))$ | New radical generator |
|---|---|---|---|
| 1 | 60 | 15 | $\sqrt{15} \in K_5$ |
| 6 | 1260 | 35 | $\sqrt{35} = \sqrt{5}\sqrt{7} \in K_7$ |
| 10 | 3300 | 33 | $\sqrt{33} = \sqrt{3}\sqrt{11} \in K_{11}$ |
| 12 | 4680 | 130 | $\sqrt{130} = \sqrt{2}\sqrt{5}\sqrt{13} \in K_{13}$ |
| 16 | 8160 | 510 | $\sqrt{510} = \sqrt{2}\sqrt{3}\sqrt{5}\sqrt{17} \in K_{17}$ |
| 18 | 10260 | 285 | $\sqrt{285} = \sqrt{3}\sqrt{5}\sqrt{19} \in K_{19}$ |
| 22 | 15180 | 345 | $\sqrt{345} = \sqrt{3}\sqrt{5}\sqrt{23} \in K_{23}$ |
| 28 | 24360 | 870 | $\sqrt{870} = \sqrt{2}\sqrt{3}\sqrt{5}\sqrt{29} \in K_{29}$ |

Every prime $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$ enters the radical field by $k = p - 1 \leq 28$, consistent with T4 Theorem 5.1's prime-rung criterion (T0 Theorem 5.1).

**Cyclotomic content.** The cyclotomic part lives in $\mathbb{Q}(\zeta_{30})$, of degree $\varphi(30) = 8$ over $\mathbb{Q}$. CRT decomposes the character index $c \in \mathbb{Z}/30$ via $\mathbb{Z}/30 \cong \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$, so each character factorizes as $\chi_c = \chi_{c_2}^{(2)} \otimes \chi_{c_3}^{(3)} \otimes \chi_{c_5}^{(5)}$ on the CRT-decomposed labels. The conductor structure has packets at every divisor $d \in \{2, 3, 5, 6, 10, 15, 30\}$ with $|C_d| = \varphi(d)$:

| $d$ | $\varphi(d)$ | Packet contents (representative) |
|---|---|---|
| 2 | 1 | $c = 15$ |
| 3 | 2 | $c = 10, 20$ |
| 5 | 4 | $c = 6, 12, 18, 24$ |
| 6 | 2 | $c = 5, 25$ |
| 10 | 4 | $c = 3, 9, 21, 27$ |
| 15 | 8 | $c = 2, 4, 8, 14, 16, 22, 26, 28$ |
| 30 | 8 | $c = 1, 7, 11, 13, 17, 19, 23, 29$ |

Total: $1 + 2 + 4 + 2 + 4 + 8 + 8 = 29 = N - 1$ as required.

**Arithmetic field.** $K_{30} \cdot \mathbb{Q}(\zeta_{30})$: $K_{30}$ has degree $2^{10} = 1024$; $\mathbb{Q}(\zeta_{30})$ has degree $\varphi(30) = 8$ and contains $\sqrt{5}$ (since $5 \equiv 1 \pmod 4$, the Gauss-sum identity places $\sqrt{5} \in \mathbb{Q}(\zeta_5) \subset \mathbb{Q}(\zeta_{30})$); intersection $K_{30} \cap \mathbb{Q}(\zeta_{30}) = \mathbb{Q}(\sqrt{5})$ of degree 2. Hence

$$[K_{30} \cdot \mathbb{Q}(\zeta_{30}) : \mathbb{Q}] = \frac{1024 \cdot 8}{2} = 4096.$$

**Structural features visible at $N = 30$.** The $29 \times 29$ matrix $U_{30}$ exhibits all three structural features of T9 simultaneously: - *Radical-cyclotomic factorization* (Cor 2.4): each entry $U_{kc} = R_k \cdot C_{kc}$. - *Conductor-block columns* (§4.1): seven column groups of sizes $(1, 2, 4, 2, 4, 8, 8)$ — the richest conductor stratification of any worked example in §5. - *Last-row uniformity* (Prop 3.1): $U_{29,c} = -\omega^{29c}/\sqrt{29}$ with constant modulus $1/\sqrt{29}$.

The full conductor-block decomposition of $U_{30}$ requires the Gauss-sum decomposition forward door (§7.4), since the three distinct primes $\{2, 3, 5\}$ produce non-trivial cross-prime interaction at the cubic Vandermonde level. $N = 30$ is the natural next benchmark after the $N = 6$ worked example (§5.2): both are squarefree-composite, but $N = 30$ has three prime axes versus $N = 6$'s two, so the cross-prime ANOVA (T0 §4.4) is structurally richer.

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## 6. Connection to standard transforms

### 6.1 Helmert vs. DFT

The character basis $\{\tilde{\chi}_c\}$ is the **discrete Fourier transform (DFT)** basis on $\mathbb{Z}/N$, with the trivial character $\chi_0$ removed. The transfer matrix $U$ is therefore the change-of-basis from the Helmert basis to (the non-trivial part of) the DFT basis.

The Helmert basis is itself a **classical contrast basis** in analysis of variance (ANOVA) and orthogonal polynomial regression: each $h_k$ is the $k$-th order discrete orthogonal contrast.

### 6.2 Helmert vs. DCT and DST

The discrete cosine and sine transforms (DCT, DST) of standard signal processing are real-valued analogs of the DFT, with various boundary conditions. The Helmert basis is qualitatively different — it is *integer* rather than *trigonometric*, and its structural feature is the radical normalization rather than orthogonal cosines/sines.

The trigonometric form of $U_{kc}$ (Theorem 2.2) makes manifest that the transfer matrix has **DCT/DST-flavored** sine envelopes $\sin(k\theta_c)/\sin\theta_c$, but the integer Helmert structure remains visible through the radical normalizers $1/\sqrt{Nk(k+1)}$.

### 6.3 Helmert vs. Walsh-Hadamard

For power-of-two $N = 2^a$, the **Walsh-Hadamard transform** uses $\pm 1$-valued basis vectors of $\mathbb{R}^N$ structured by binary expansion. The Helmert basis is *not* the same as the Walsh-Hadamard basis: Helmert is triangular (each $h_k$ supported on positions $0, \ldots, k$), while Walsh-Hadamard is fully spread.

A future paper (T9.3, planned) could explore the precise relationship at power-of-two $N$, including whether the Helmert and Walsh-Hadamard bases generate the same algebraic group action on $H_N$.

## 7. Forward doors

### 7.1 Complex extension via T7

The Helmert basis was developed for real ledgers; in the complex amplitude setting of T7, the natural extension uses Hermitian Helmert contrasts on $\mathbb{C}^N$. The complex transfer matrix $U_{\mathbb{C}}$ on $H_N \otimes \mathbb{C}$ is identical to $U$ as defined here (since the character basis is already complex), but its action on T7's complex apex generator $\mathcal{M}_z$ introduces new structure.

Forward paper title (planned): **T9.1 — The Complex Helmert-Fourier Transfer Matrix.**

### 7.2 Multiplicative extension via T8

T8 introduced the multiplicative-character basis $\{\psi\}$ of $\widehat{(\mathbb{Z}/N)^\times}$ on units. The natural **multiplicative-Helmert transfer matrix** $V$ would relate a Helmert-like basis on units (suitably constructed) to the multiplicative character basis. The Helmert basis itself does not naturally restrict to units; an analog construction on $(\mathbb{Z}/N)^\times$ requires defining "discrete contrasts" indexed by the unit group structure.

Forward paper title (planned): **T9.2 — The Multiplicative Helmert Basis on Units.**

### 7.3 Walsh-Hadamard comparison at $N = 2^a$

For $N = 2^a$, both the Helmert and Walsh-Hadamard bases are integer-rational orthogonal bases of $H_N$. Their precise algebraic relationship — including the change-of-basis matrix between them, and whether they coincide in any natural sense — is a forward door.

Forward paper title (planned): **T9.3 — Helmert vs Walsh-Hadamard at Power-of-Two $N$.**

### 7.4 Full Gauss-sum decomposition of conductor blocks

The conductor-block columns $U^{(d)}$ of §4 contain Gauss-sum-flavored content that connects to T8's multiplicative-Fourier framework. The precise Gauss-sum decomposition of each column block — expressing $U^{(d)}$ as a structured product of Gauss sums and unit radicals — is the natural meeting point of T9 and T8.

Forward paper title (planned): **T9.4 — Gauss-Sum Decomposition of the Conductor Blocks.**

### 7.5 Spectral and algebraic-group structure

$U \in U(N-1)$ is a specific unitary matrix; its spectrum (eigenvalues and eigenvectors) carries arithmetic information. Computing the spectrum of $U$ at small $N$ and connecting it to known number-theoretic quantities (Gauss-sum traces, regulator-type invariants of $K_N$, etc.) is a forward door for the analytic side of T9.

Forward paper title (planned): **T9.5 — Spectral Theory of the Helmert-Fourier Transfer Matrix.**

## 8. Coda

The Helmert-Fourier transfer matrix $U$ is the apex's arithmetic Jacobian between two distinct orthogonal bases of $H_N$:

- The **Helmert basis** $\{e_k\}$, integer-rational, carrying the **dimensional axis** of the apex (T4 dimensional ladder and its multiquadratic radical clock).
- The **character basis** $\{\tilde{\chi}_c\}$, cyclotomic-complex, carrying the **conductor axis** of the apex (T0/T3 conductor packets and their additive Fourier structure).

Every computation in the apex framework that crosses these two axes — every transfer between dimensional content (Helmert order $k$) and conductor content (character index $c$) — passes through $U$. The closed-form entries (Theorem 2.1, equivalent forms in §2.2, §2.3), the radical-cyclotomic factorization (Corollary 2.4), and the row/column structural propositions (§3) together provide the entry-level structure theorem; the conductor-block structure (§4) and worked examples (§5) exhibit the apex's arithmetic Jacobian explicitly for small $N$.

The one-sentence apex of T9:

> **The Helmert-Fourier transfer matrix $U_{kc} = R_k \cdot C_{kc}$ factors entry-wise into a radical $R_k = 1/\sqrt{Nk(k+1)}$ in the multiquadratic field $K_N$ (the Helmert axis content) and a cyclotomic polynomial $C_{kc}$ in $\omega^c$ over $\mathbb{Q}(\zeta_N)$ (the conductor axis content), uniting the dimensional and conductor arithmetics of the categorical-simplex apex into a single unitary matrix in $U(N-1)$ over the compositum $K_N \cdot \mathbb{Q}(\zeta_N)$.**

And the closing slogan:

> **Two orthogonal bases of $H_N$: one integer-rational, one cyclotomic. Their transfer matrix is the apex's arithmetic Jacobian.**

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